ARVR REPORT

1. Topic of Mini Project

Title: AR Cooking with Interactive Guide

2. Description

2.1 Introduction

Augmented Reality is a technology that overlays virtual objects onto the real world through a mobile camera. It helps users interact with digital content in a more immersive and engaging way.

Our mini project, **AR Cooking with Interactive Guide**, is an AR-based cooking assistant that helps users learn recipes in an interactive manner. The application places a virtual rat chef on a real-world surface such as a table or kitchen counter. The rat then guides the user through recipe selection, ingredient checking, cooking steps, and finally displays the completed dish in AR.

The project is inspired by the idea of making cooking more visual, fun, and guided rather than showing only plain text instructions like traditional cooking apps.

2.2 Problem Statement

Traditional recipe applications usually provide only text or video instructions. These methods are helpful, but they are not interactive enough for beginners. Users may still find it difficult to visualize the process, remember ingredients, or follow the steps properly.

There is a need for a more engaging solution where users can interact with cooking guidance directly in their real environment.

2.3 Objective

The main objectives of this project are:

- To develop an AR-based cooking guide using Unity and AR Foundation
- To detect real-world surfaces and place a virtual chef on them
- To allow the user to select a recipe from the app
- To provide an ingredient checklist before cooking begins
- To guide the user with step-by-step cooking instructions
- To show the final dish in AR after the recipe is completed
- To make the cooking experience interactive and visually appealing

2.4 Scope of the Project

This project focuses on creating a mobile AR cooking guide for a small number of recipes. The application currently supports dishes such as:

- Pasta
- Omelette
- Sandwich
- Pancakes

The system is designed as a prototype to demonstrate how AR can improve recipe learning and interaction.

2.5 System Overview (Working)

The application works in the following steps:

Step 1 — AR Initialization

The camera opens and detects flat surfaces using AR technology.

Step 2 — Surface Detection

Surfaces such as tables or counters are detected and highlighted.

Step 3 — Object Placement

User taps the surface → the **rat chef appears**.

Step 4 — UI Activation

After placement:

- “Recipes” button appears
- Rat greets user

Step 5 — Recipe Selection

User selects a dish:

- Pasta
- Omelette
- Sandwich
- Pancakes

Step 6 — Ingredient Checklist

User checks required ingredients
“Ingredients Ready” activates only after all are selected

Step 7 — Cooking Steps

User follows:

- Step-by-step instructions
- Next / Previous navigation

Step 8 — Final Dish Display

After completion:

- Dish appears in AR
- Floating animated model

Step 9 — End Sequence

- Screen fades to black
- App resets

3. Tools Used

Tool	Purpose
Unity	**Main development engine**
AR Foundation	**AR functionality**
ARCore	**Android AR support**
C# (Visual Studio)	**Scripting**
Blender	**3D model editing**
TextMeshPro	**UI text rendering**
Android Build Tools	**App deployment**

4. Sample Code

4.1 Object Placement (AR Raycast)

C#

```
if (raycastManager.Raycast(touch.position, hits, TrackableType.PlaneWithinPolygon))
{
   Pose hitPose = hits[0].pose;

   if (spawnedObject == null)
   {
      spawnedObject = Instantiate(objectToPlace, hitPose.position, hitPose.rotation);
   }
}
```

4.2 Ingredient Validation

C#

```
foreach (Toggle toggle in ingredientToggles)
{
   if (!toggle.isOn)
   {
      allChecked = false;
      break;
   }
}
ingredientsReadyButton.interactable = allChecked;
```

4.3 Floating Dish Animation

C#

```
float y = startPos.y + Mathf.Sin(Time.time * floatSpeed) * floatHeight;
transform.position = new Vector3(transform.position.x, y, transform.position.z);
transform.Rotate(0, rotateSpeed * Time.deltaTime, 0);
```

4.4 Fade Screen Effect

C#

```
fadeImage.color = new Color(0, 0, 0, alpha);
```

5. Modules of the Project

5.1 AR Surface Detection Module

This module uses AR Foundation and ARCore to detect flat surfaces in the real environment. These surfaces are visualized using plane detection.

5.2 Object Placement Module

This module allows the user to place the rat chef on the detected surface by tapping the screen.

5.3 Recipe Selection Module

This module provides the list of available dishes for the user to choose from.

5.4 Ingredient Checklist Module

This module displays the ingredients required for the chosen dish. The user marks all required items before beginning the cooking process.

5.5 Cooking Guidance Module

This module shows the step-by-step cooking instructions and allows navigation using Start, Next, Previous, and Reset buttons.

5.6 Final Dish Display Module

After completion of the recipe, the final dish model appears in AR with floating animation.

5.7 UI and Interaction Module

This module handles text panels, buttons, speech bubbles, transitions, and overall user interaction.

6. Screenshots

1. AR Surface Detection

2. Rat Placement

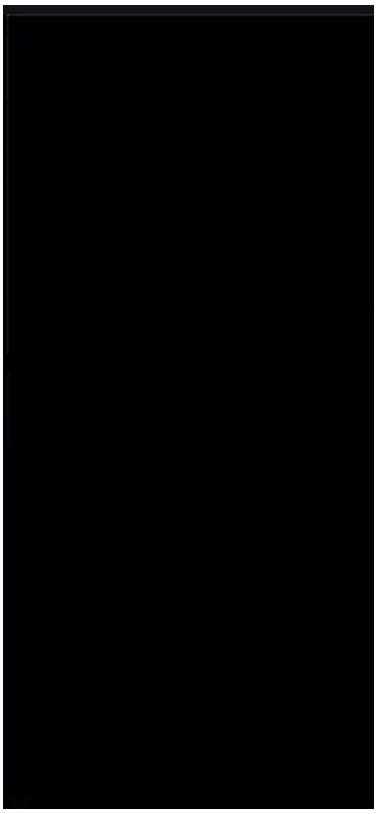
3. Recipe Selection Panel

4. Ingredient Checklist

5. Cooking Steps Panel

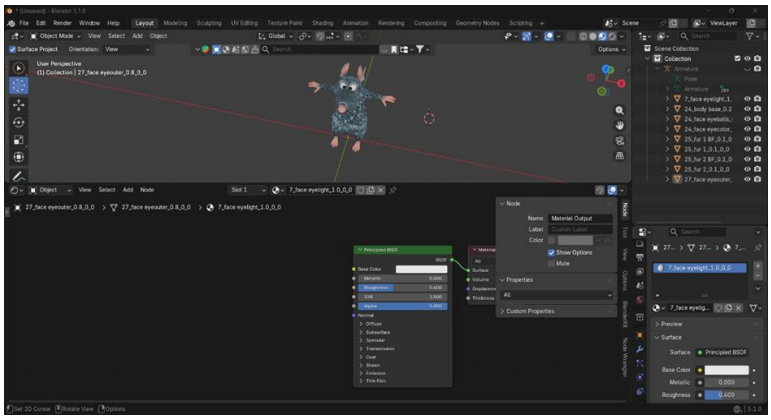
6. Final Dish Output

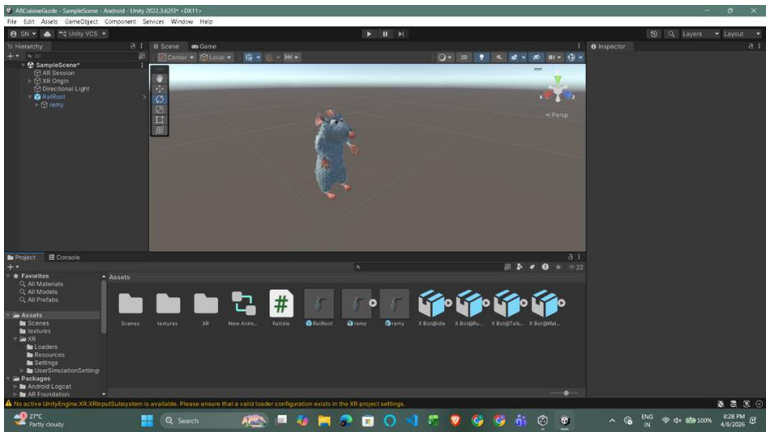
7. Fade Screen

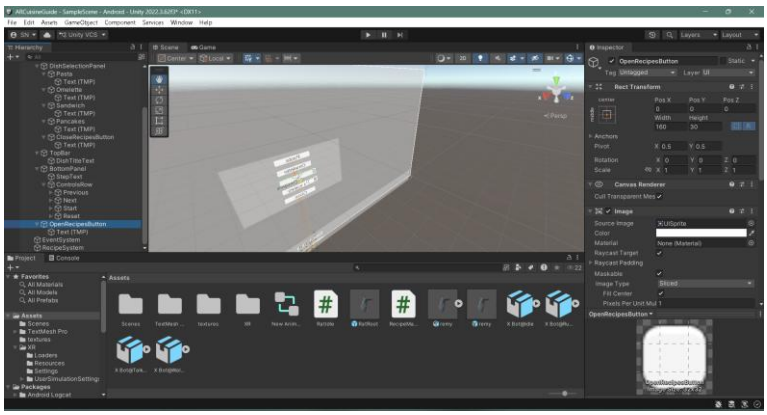


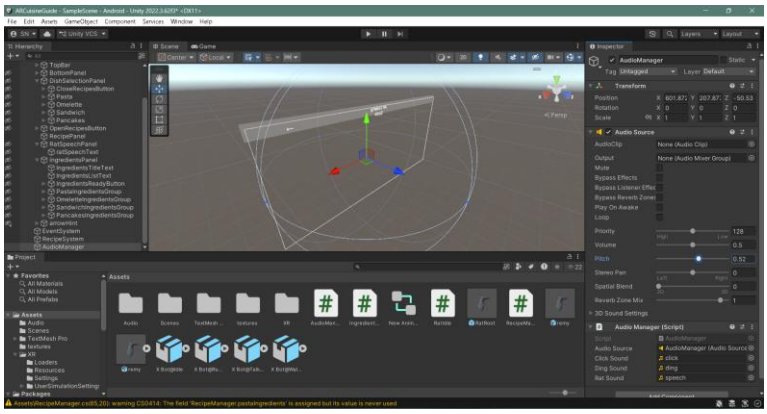
8. Reset Flow

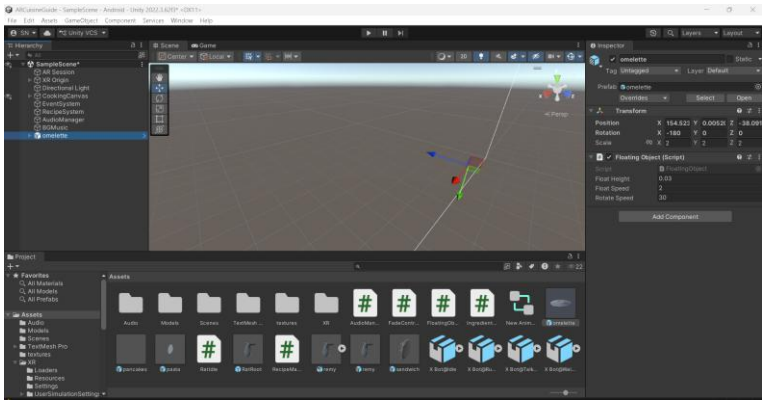
9. Unity, Blender and VSCode

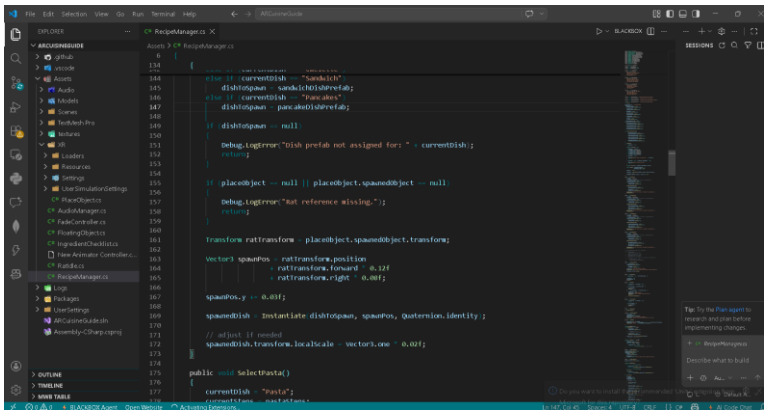












8. Conclusion

This project successfully demonstrates how Augmented Reality can be used to create an interactive cooking assistant. By combining AR visualization, UI interaction, and guided instructions, the system provides a more engaging and user-friendly experience compared to traditional recipe applications.

The project enhances learning through:

- visual interaction
- step-by-step guidance
- real-world integration